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Python for Data Science
Module 4

Recap, Python through data, intro to pandas

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About me

Alberto Cámara

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- **PhD in Mathematics**
- Work as a **Data Freelancer**
- **PyBcn** organiser
- Plenty of experience leading software developer teams, notably in **Data Science**, **Machine Learning** and **AI**

I take over from here: **modules 4–8, the hackathon and the final exam**.



There are no stupid questions

- I am very happy to take questions mid-lecture
- Worst thing that can happen is that I ask to leave the question for the end of the lecture

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Where we are

Module 0 · Terminal essentials

Just enough terminal to know where you are, find your project, and run it.

```
pwd              # where am I?
ls               # what is here?
cd retail-analysis
conda activate pds
python hello.py
```

When a command fails, first ask: **where am I?** And: **which Python?**

Module 1 · From notebook to project

```
retail-analysis/
  README.md
  environment.yml
  analysis.ipynb
  data/
    raw/
  src/
```

- A project has a **home**: run from its root
- **Relative paths** belong to the project, absolute paths to one machine
- Original inputs live in `data/raw/`, apart from code

Module 2 · Packaging and testing

```python
from retail_analysis.prepare_data import normalize_ids


def test_normalize_ids_normalizes_case():
    ids = pd.Series(["c001"])
    assert normalize_ids(ids).tolist() == ["C001"]
```

- `pyproject.toml` describes the project
- `pip install -e .` makes `src/` importable
- `pytest` keeps expected behaviour intact as code changes

Checkpoint 02

Module 3 · Git and collaboration

- **Save** a file → **commit** a snapshot → **push** to share it
- A useful commit message tells a story: *"Document how to run tests"*
- Git is the tool, GitHub is the host; you used GitHub Desktop in pairs

Checkpoint 03

You'll use exactly this today: **clone** the course repository and **pull** every week.

Where the course goes

Date	Module
Wed 4 Nov	M04 · Intro to pandas
Wed 11 Nov	M05 · Grain, group-by, joins
Wed 18 Nov	M06 · Visualisation with plotly
Wed 25 Nov, 9:00	M07 · Better charts, maps
Wed 25 Nov, 14:00	M08 · End-to-end case + exam prep
Wed 2 Dec	*Hackathon* (pending coordination with the AI course)
Wed 16 Dec	**Final exam**, 8:45, room 3A01

Assessment

Component	Weight
Checkpoints: CP02, CP03, CP-pandas, CP-viz (5% each)	**20%**
Hackathon: our data-preparation part 20 points + the AI part 10	**30%**
Final exam: five short written questions	**50%**

- **CP-pandas**: out 11 Nov, due **Tue 17 Nov** 23:59
- **CP-viz**: out 25 Nov, due **Tue 1 Dec** 23:59
- Each checkpoint: three items, each *met* or *not met*, one notebook that runs top to bottom
- **Attendance is compulsory**

What the exam asks

Five short questions, one per topic. Reasoning, not syntax.

Topic	Where you learn it
Reproducibility	M00–M03 project structure · M08
Testing and validation	M02 pytest · M05 assertions
Layered data architecture (raw → clean → mart)	M04 · M05 · M08
Grain and joins	M05
Efficient, tool-appropriate workflows	M05 group-by instead of loops · M06–M07 chart choice

The thread through every module: **how do you know this number is right?**

One project, all course long

The question

Do late deliveries hurt review scores, and where should operations act first?

Real data from **Olist**, a Brazilian marketplace: ~99,000 orders, nine tables.

Every lab builds one piece. In M08 we put it together.

The repository

`github.com/ber2/2026-esade-pds-course-materials`

- `sessions/m04/`: today's material (**never edit**)
- `work/m04/`: your copies (yours to edit)
- `data/raw/`: the Olist CSVs

Each session's results are published at the start of the next, so nobody gets stuck.

Let's open the project you built

`retail-analysis`, as you left it after Module 3.

1. The layout: `data/raw/`, `src/retail_analysis/`, `tests/`
2. `python -m pytest`: three green tests
3. `analysis.ipynb`: the revenue number

The tests pass. Is the number right?

```
orders = orders[orders["status"] == "completed"]
len(orders)                    # 108
orders = orders.dropna()
len(orders)                    # 102
```

Six completed orders vanished. Which six? Do we care?

All six are missing `shipping_country`, a column the analysis **never uses**.
Their revenue is gone from the total, and nothing warned us.

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Python through data

Lists, dictionaries, loops, functions

Who is in the room?

Hands up:

1. Who did the **Data Science Pre-Program**?
2. Who **finished the warm-up** notebook?
3. Who got **stuck** somewhere in it?

A table is a list of dictionaries

```
orders = [
    {"order_id": "o1", "state": "SP", "price": 129.90, "delivered": 9},
    {"order_id": "o2", "state": "RJ", "price": 49.90, "delivered": 20},
    {"order_id": "o4", "state": "MG", "price": 199.00, "delivered": None},
]
```

- **list**: rows, in order, by position: `orders[0]`
- **dictionary**: one row, fields by name: `orders[0]["price"]`
- **loop**: do something for every row
- **function**: name a rule once: `is_late(order)`

Demo: `demo.ipynb`, section 1.

From loops to pandas

Python

```
total_revenue = 0
for order in orders:
    total_revenue += order["price"]
```

pandas

```
df = pd.DataFrame(orders)
df["price"].sum()
```

A DataFrame is an **object**:

- **attributes** are data you read: `df.shape`, `df.columns`
- **methods** are actions you call: `df.head()`, `df.sum()`

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Meet Olist

The business

- Olist connects **small shops** to the big Brazilian online stores
- A customer orders, a **seller** ships, the customer gets an **estimated delivery date**, and later leaves a **review**
- Our client: the **marketplace operations team**

Orders	99,441
Period	Sep 2016 – Oct 2018
Tables	9
Customers' states	27

Nine tables, one question each

Table	One row is…
orders	an order
order_items	an item line in an order
order_payments	a payment of an order
order_reviews	a review
customers	a customer *as recorded on one order*
products · sellers	a product · a seller
category translation	a category (Portuguese → English)

Today: **orders**. Lab: **customers** and **products**.

First look: five methods

```
orders = pd.read_csv(RAW / "olist_orders_dataset.csv")
orders.shape        # (99441, 8)
orders.head()
orders.info()       # dtypes and non-null counts
orders.describe()
orders.dtypes
```

All five date columns come in as `object`: text. pandas does not guess dates.

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Types and selection

Parse dates when you read

```
orders = pd.read_csv(RAW / "olist_orders_dataset.csv", parse_dates=DATE_COLUMNS)
```

You want	Write
one column	`orders["order_status"]`
several columns	`orders[["order_id", "order_status"]]` (a **list**)
rows and columns by label	`orders.loc[0:2, ["order_id", "order_status"]]`
rows that meet a condition	`orders[orders["order_status"] == "delivered"]`
one of several values	`orders["order_status"].isin(["canceled", "unavailable"])`
a range	`orders["order_purchase_timestamp"].between("2018-01-01", "2018-02-01")`
how often each value appears	`orders["order_status"].value_counts()`

Your turn: Black Friday week

Black Friday week 2017 is **20 to 26 November**. What goes in the `...`?

```
black_friday_week = orders["order_purchase_timestamp"].between(...)
black_friday_week.sum()
```

Missing values are information

2,965 orders have no delivery date

- Shipped, canceled, unavailable, invoiced… were **never delivered**: the `NaN` *is* the status
- Only **8** "delivered" orders lack a date

	Rows
all orders	99,441
after `orders.dropna()`	96,461
delivered, with a delivery date	96,470

`dropna(subset=[...])` says **which** column must be present. Blanket `dropna()` decides for you.

Derived columns, and a number to defend

`delivery_days` = delivery date − purchase date, in days. Median: **10 days**.

How often are we late?

Definition	Late
delivery timestamp > estimated date	8.1%
delivery **day** > estimated date	**6.8%**

Delivered at 15:00 on the estimated day: late? Both versions run. Only the business can decide.

Lab: customers and products (20 min)

`sessions/m04/lab.ipynb` → copy to `work/m04/lab.ipynb`

- **Part 0**: Python on 20 real item lines (0d is the stretch)
- **Part 1**: customers. A zip code that lost its leading zero
- **Part 2**: products. Typos in column names, missing categories, products that weigh 0 g
- Each exercise has a **check cell**: silence means correct

Next week: what is one row?

- Save the typed tables to the **clean** layer (`data/clean/`, parquet)
- **Grain**: one row of each table is… what, exactly?
- Group-by, and the join that silently adds **4 million BRL** of revenue
- **CP-pandas** is released

Before then: finish the lab, and keep `work/m04/`.

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See you next Wednesday

Everything is in the repository. Pull before class.

Bring questions.